WORKED SOLUTION

STEP II 1990, Question 6

Pure • Integer Inequalities and Fractions

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q6

The key idea

Everything rests on one fact about integers: a strict inequality between integers can be sharpened by 1. If $a < b$ then $b \geq a + 1$, and if $d < c$ then $d \leq c - 1$. That turns (ii) into a one-line estimate. The later parts are then (ii) applied to cleverly chosen integers: clear the denominators in the fractional inequalities, and feed the resulting integer inequalities back into (ii).

Solution

Throughout, all letters stand for positive integers.

Part (i)

Since $b > a \geq 1$ and b is an integer, $b \geq 2$. Multiplying by $c > 0$ gives $bc \geq 2c$.

Since $c > 1$ and c is an integer, $c \geq 2$, so $2c = c + c \geq c + 2$. Together, $bc \geq 2c \geq 2 + c$, as required.

Part (ii)

Because the inequalities are strict and the numbers are integers, $b \geq a + 1$ and $d \leq c - 1$. Using $c > 0$ in the first and $a > 0$ in the second,

$$\begin{aligned} bc - ad &\geq (a+1)c - a(c-1) \\ &= ac + c - ac + a \\ &= a + c. \end{aligned}$$

So $bc - ad \geq a + c$, with equality exactly when $b = a + 1$ and $d = c - 1$.

(This is the same "sharpen by 1" trick as in (i), which is a warm-up; we won't need (i) again.)

Part (iii)

Multiply out the two inequalities, using $b, d > 0$:

$$\frac{a}{b} < p \iff a < bp, \qquad p < \frac{c}{d} \iff dp < c.$$

Now apply (ii) with b replaced by bp and d replaced by dp (both positive integers). The hypotheses $a < bp$ and $dp < c$ are exactly what we have, so

$$(bp)c - a(dp) \geq a + c, \qquad \text{that is,} \qquad (bc - ad)p \geq a + c.$$

Part (iv)

Clearing denominators (all positive) gives two integer inequalities:

$$aq < bp \qquad \text{and} \qquad dp < cq.$$

Also $\frac{a}{b} < \frac{c}{d}$, so $ad < bc$ and $bc - ad$ is a *positive* integer; we may therefore divide by it without reversing any inequality.

? **Ask yourself** Which four numbers should play the roles of a, b, c, d in (ii)?

We have two strict inequalities, and (ii) needs exactly two: "first $<$ second" and "fourth $<$ third". There are two natural ways to match them up, and each gives one of the bounds.

The bound on p. Take $aq < bp$ as the "$a < b$" inequality and $dp < cq$ as the "$d < c$" one. Part (ii), applied to aq, bp, cq, dp, gives

$$(bp)(cq) - (aq)(dp) \geq aq + cq, \qquad \text{so} \qquad pq(bc - ad) \geq q(a + c).$$

Dividing by $q > 0$ and then by $bc - ad > 0$,

$$p \geq \frac{a + c}{bc - ad}.$$

The bound on q. Now swap the roles: take $dp < cq$ as the "$a < b$" inequality and $aq < bp$ as the "$d < c$" one. Part (ii), applied to dp, cq, bp, aq, gives

$$(cq)(bp) - (dp)(aq) \geq dp + bp, \qquad \text{so} \qquad pq(bc - ad) \geq p(b + d).$$

Dividing by $p > 0$ and by $bc - ad > 0$,

$$q \geq \frac{b + d}{bc - ad}.$$

Both bounds are proved. (Part (iii) is the special case $q = 1$ of the first one.)

The fractions between $8/9$ and $9/10$

Take $a = 8$, $b = 9$, $c = 9$, $d = 10$. Then

$$bc - ad = 81 - 80 = 1,$$

so any fraction p/q (with p, q positive integers, in lowest terms or not) lying strictly between $8/9$ and $9/10$ satisfies

$$p \geq \frac{8+9}{1} = 17, \qquad q \geq \frac{9+10}{1} = 19.$$

A denominator less than 20 must therefore be exactly $q = 19$. Then we need

$$\frac{8}{9} < \frac{p}{19} < \frac{9}{10} \iff \frac{152}{9} < p < \frac{171}{10} \iff 16.88\ldots < p < 17.1,$$

so $p = 17$. And $17/19$ really does lie in the interval, since $8 \cdot 19 = 152 < 153 = 9 \cdot 17$ and $17 \cdot 10 = 170 < 171 = 9 \cdot 19$.

Hence the only fraction with denominator less than 20 between $8/9$ and $9/10$ is

$$\frac{17}{19}.$$

Because the bound $q \geq 19$ applies to every way of writing a fraction, not only to lowest terms, no non-reduced form like $34/38$ sneaks in with a smaller denominator either.

Checks

- $17/19$ is the *mediant* $\frac{8+9}{9+10}$, and both bounds in (iv) are attained by it, which is a good sign the bounds are sharp.
- Decimals: $8/9 \approx 0.8889$, $17/19 \approx 0.8947$, $9/10 = 0.9$.
- We tested (i) and (ii) for all values up to 29, tested (iii) and (iv) on about 60,000 random cases, and listed every p/q with $q \leq 19$ between $8/9$ and $9/10$ by brute force: only $17/19$ appears.

Common slips

- Forgetting to say that $bc - ad > 0$ before dividing by it in (iv).
- Using only the p bound in the final part. It allows q up to 19 but doesn't by itself rule out smaller denominators; the q bound is what does the work.
- Getting only one bound in (iv) from (ii): the second needs the roles of the two inequalities swapped.

Beyond STEP

Two fractions $a/b < c/d$ with $bc - ad = 1$ are called *Farey neighbours*, and this question proves the key fact about them: the fraction with the smallest denominator between them is their mediant $\frac{a+c}{b+d}$. Repeatedly inserting mediants builds the Farey sequences and the Stern–Brocot tree, which list every positive rational exactly once and are closely tied to continued fractions and the best rational approximations of real numbers.

Davenport's *The Higher Arithmetic* is a lovely place to read about continued fractions and approximation.

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